Smart Hospital Queue Management App

Design Thinking for Innovation — Summative Assessment

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1. Project Overview

This project focuses on designing a smart hospital queue management app using the Design Thinking approach. The goal is to reduce patient stress, provide real-time queue updates, and make waiting more comfortable and engaging. The app targets patients, caregivers, and hospital staff across all age groups and health conditions.

2. Empathize & Define

Key Patient Problems in Waiting Areas

1. Uncertainty about queue position and wait time causes anxiety and frustration.
2. High anxiety and stress — especially for elderly, children, and chronic illness patients.
3. Boredom and lack of engagement; patients feel disconnected during long waits.
4. Physical discomfort from poor seating, noise, and lack of basic amenities.
5. Communication barriers for hearing/vision-impaired or non-native speaking patients.
6. No personalized support for mobility issues, emotional needs, or language preferences.

User Personas

- Meera, 65 — Senior patient, hard of hearing; needs silent visual notifications.
- Anika, 8 — Child patient; easily bored, prefers interactive games and short videos.
- Rajesh, 35 — First-time visitor; nervous and overwhelmed, needs clear guidance.
- Mrs. Desai, 58 — Chronic illness patient; stressed about delays, needs calm environment.
- Amit, 42 — Working professional; impatient, needs accurate wait time predictions.
- Priya, 28 — Caregiver accompanying elderly mother; needs updates while multitasking.
- Omar, 72 — Mobility-challenged senior; needs accessible seating and staff support.
- Isha, 24 — Anxiety-prone patient; needs breathing exercises and frequent reassurance.

Problem Statement

"How might we transform hospital waiting from a stressful, uncertain experience into a calm, engaging, and informative journey that makes patients feel supported and valued?"

3. Ideation — Key Features

1. Queue Progress Tracker

Shows real-time queue position, estimated wait time, and a progress bar. Large, bold numbers reduce uncertainty. Updates every 2–3 minutes. Color-coded progress bar (green = close to turn).

2. Distraction Hub

Provides calming videos (nature walks, guided tours, animal videos), mini games (puzzles, memory match), and audio content (meditation, podcasts, calm music). Organized by category for easy browsing.

3. Silent Notifications

Sends gentle vibration alerts and in-app banners when turn is approaching — no loud announcements. Alerts at 5 minutes and 2 minutes before turn. Optional SMS reminders.

4. Mood-Based Calming Support

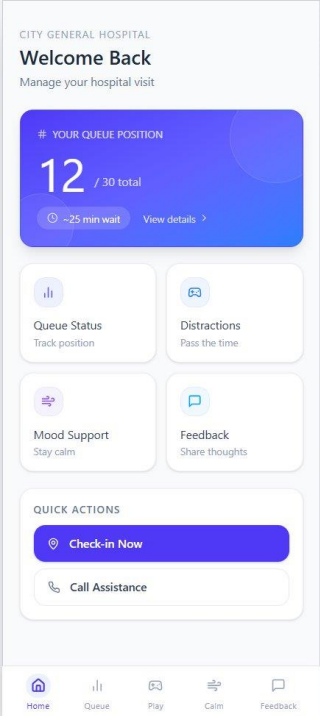
Mood check-in via emoji scale. App suggests activities based on stress level: breathing exercises for anxiety, games for boredom, calm sounds for restlessness. Includes affirmations and art therapy.

5. Feedback System

Star ratings, open-ended suggestions, and checkboxes for specific areas (Wait Time Accuracy, App Navigation, Distraction Features, Overall Service). Anonymous submission option included.

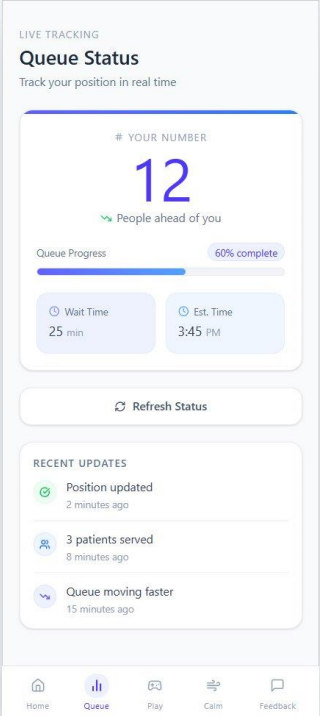
4. Prototype — App Screens

The following 5 screens were designed in Figma using a clean, modern interface with an indigo/blue color palette, Poppins headings, and Inter body text.



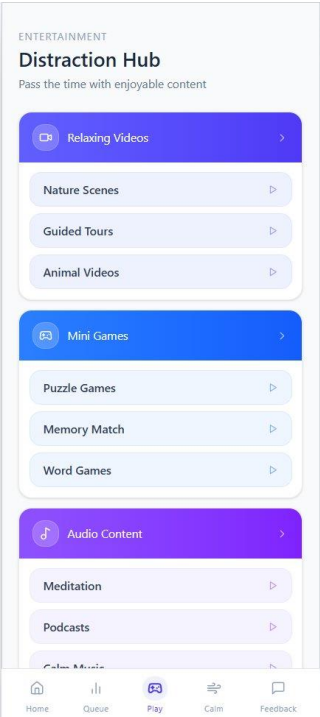
Screen 1: Home / Dashboard

Shows queue position #12/30, ~25 min wait. Quick access to all features and Check-in button.



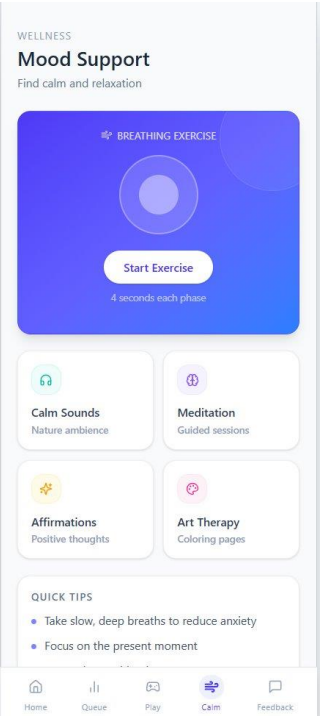
Screen 2: Queue Status

Live tracking with progress bar (60%), wait time (25 min), estimated time (3:45 PM), and recent updates.



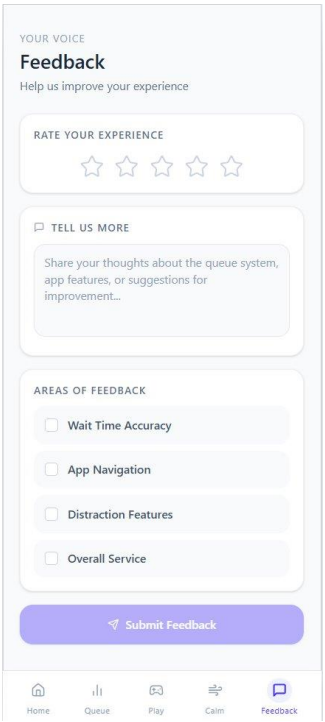
Screen 3: Distraction Hub

Categorised content: Relaxing Videos, Mini Games, and Audio Content with expandable sub-items.



Screen 4: Mood Support

Breathing exercise animation, Calm Sounds, Meditation, Affirmations, Art Therapy, and Quick Tips.

 **Screen 5: Feedback** Star rating, open text field, and checkbox areas (Wait Time, App Navigation, etc.) with Submit button.	

5. Testing & User Feedback

Testing Method

The prototype was shared with 5 users (students, a working professional, and a senior) who were asked to complete 3 tasks: check queue position, use the distraction hub, and submit feedback.

Feedback Received

- Interface is simple, clean, and easy to navigate.
- Queue tracking feature was found most helpful — reduces stress immediately.
- Users suggested larger font size for elderly users.
- Requested more game variety in the Distraction Hub.

Improvements Made

- Increased font size across all screens for better readability.
- Improved button contrast and visibility.
- Added more content options in the Distraction Hub (Affirmations, Art Therapy).

6. Conclusion

The CareHub prototype successfully reimagines the hospital waiting experience by making it calmer, more informed, and engaging. The Design Thinking process helped identify real user pain points and translate them into practical app features that are inclusive, accessible, and empathetic.

7. GitHub Repository

Repository Link	https://github.com/tanmaythakur09/IDAI106_1000389-TanmayThakur
Repository Name	IDAI106_1000389-TanmayThakur

8. Tools Used

- Figma — App screen design and prototype
- Miro / MindMeister — Empathy maps and ideation
- GitHub — Repository and documentation
- ChatGPT / Claude — Research support and content planning

9. Design System

Primary Color	#4F46E5 — Indigo
Secondary Color	#06B6D4 — Cyan
Background	#F1F5F9 — Light Gray
Heading Font	Poppins
Body Font	Inter
UI Style	Modern, clean, rounded corners, soft shadows

10. Repository Structure

```
IDAI106_1000389-TanmayThakur/
├── Docs/
│   └── Tanmay_Thakur_1000389_Submission.docx
├── Wireframes/
│   ├── 01_Home_Dashboard.png
│   ├── 02_Queue_Status.png
│   ├── 03_Distraction_Hub.png
│   ├── 04_Mood_Support.png
│   └── 05_Feedback.png
├── Data/
│   └── user_testing_feedback.pdf
└── README.md
```

11. Submission Checklist

✓	GitHub repository created and organised
✓	8 user personas and empathy maps completed
✓	Problem statement defined
✓	All 5 compulsory features included in prototype
✓	5 app screens designed in Figma
✓	User testing conducted with 5 users
✓	Submission document uploaded to Docs/ folder
✓	Wireframe screenshots uploaded to Wireframes/ folder
✓	Access granted to ai.assignments@wacpinternational.org
✓	README.md complete with all required sections